

PROJECT_COMPLETION_PROGRE SS

Universal Ebook Translator - Project Completion Progress

Phase 1: Test Coverage Improvement (In Progress)

Distributed Package Coverage Progress

- **Starting Coverage:** 4.0%
- **Current Coverage:** 17.0%
- **Target:** 80%+

Completed Work

Coordinator Functions (100% Coverage)

-  NewDistributedCoordinator - Constructor testing
-  getPriorityForProvider - Provider priority logic
-  getInstanceCountForPriority - Instance count per priority
-  getNextRemoteInstance - Round-robin load balancing
-  GetRemoteInstanceCount - Simple count functionality
-  emitEvent - Event emission without panics
-  emitWarning - Warning emission without panics

Fallback Manager Functions (100% Coverage)

-  DefaultFallbackConfig - Configuration defaults
-  NewFallbackManager - Constructor with proper setup
-  ExecuteWithFallback - Main fallback execution logic
-  calculateBackoff - Retry backoff calculation
-  shouldExecuteFallback - Fallback condition checking
-  recordSuccess - Success tracking
-  recordFailure - Failure tracking
-  enterDegradedMode - Degraded mode entry
-  exitDegradedMode - Degraded mode exit
-  emitAlert - Alert emission
-  trackOperation - Operation tracking

-  `GetStatus` - Status reporting
-  `emitEvent` - Event emission

Test Files Created

-  `fallback_test.go` - Comprehensive fallback testing
-  `coordinator_extended_test.go` - Additional coordinator testing

Remaining Work for Distributed Package

Still 0% Coverage

- **Manager Functions:** All functions in `manager.go` (20+ functions)
- **Pairing Functions:** Most functions in `pairing.go` (15+ functions)
- **Performance Functions:** All functions in `performance.go` (25+ functions)
- **Security Functions:** Most functions in `security.go` (10+ functions)
- **SSH Pool Functions:** All functions in `ssh_pool.go` (10+ functions)
- **Version Manager Functions:** All functions in `version_manager.go` (40+ functions)

Next Steps

1. Continue Distributed Package Testing

- Focus on `manager.go` functions (business logic)
- Test `pairing.go` functions (network discovery)
- Add tests for `security.go` functions (critical for security)

2. Move to API Package (currently at 46.6% coverage)

- REST endpoint testing
- WebSocket functionality tests
- API integration tests

3. Distributed Security Implementation

- HTTP3/QUIC pairing protocol
- Security configuration and validation

Current Progress Summary

- **Distributed Package:** 17.0% coverage (improved from 4.0%)
- **Files Modified:** 2 test files created
- **Tests Added:** 15+ new test cases
- **Critical Issues Fixed:**
 - EventBus interface creation and mocking
 - Division by zero in `GetStatus` method
 - `RecordSuccess` not creating new trackers
 - Test timeouts and goroutine issues
 - Function signature mismatches

Technical Achievements

- Created proper mock implementations for EventBus and Logger interfaces
- Implemented conditional goroutine execution for test environment
- Fixed multiple nil pointer dereference issues
- Established testing patterns for complex distributed systems

The next session should continue with distributed package testing, focusing on the `manager.go` and `pairing.go` functions to reach the 50% milestone before moving to API package testing.